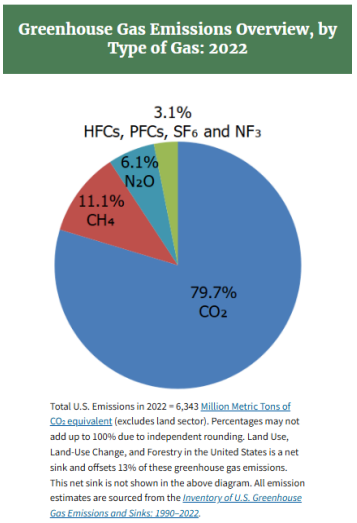
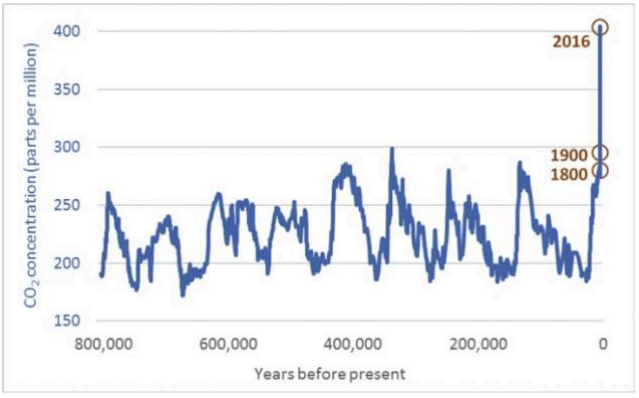
ASSIGNMENT 2 (Team - 30%)

Novel Materials Strategies to Combat Global Warming

There is overwhelming scientific evidence that the World's climate is rapidly changing. It is generally accepted that human influence has been the dominant cause of the observed global warming over the past 100 years. The extensive use of fossil fuels, land clearance for food production and increases in affluence have combined with an uncontrolled population increase to liberate increasing amounts of greenhouse gases into the atmosphere.

The figures below show that although carbon dioxide is the major contributor to global warming, other gases, such as methane (fugitive emissions from oil extraction, from agriculture/livestock and from permafrost thawing) and nitrous oxide (from agriculture) also play a major role. About 50 Gt/year of CO_2 equivalent is currently generated by human activities. See: https://www.epa.gov/ghgemissions/overview-greenhouse-gases and https://www.iea.org/geco/.





There is an urgent need to implement strategies to decrease and eventually reverse the liberation of greenhouse gases into the atmosphere. It is estimated that without new policies to mitigate climate change we can expect an increase in global mean temperature in 2100 of 3.7 to 4.8 °C, relative to pre- industrial levels. Importantly, the current trajectory of global greenhouse gas emissions is inconsistent with limiting global warming to below 2 °C, relative to pre-industrial levels. The value of 2 °C is seen to be the maximum temperature rise we can accept before there is extensive species extinction, food insecurity and social disruption. That value requires limiting CO_2 equivalent concentrations to < 450 ppm.

Efforts to control the generation of greenhouse gases are widespread in nearly every country. Efforts can include better management of population growth and energy, innovations in land use and new business models. Technologies involving developing new materials and optimal use of existing materials, have a role to play in that process. A recent publication named "Drawdown", outlined in the websites (https://drawdown.org/solutions/table-of-solutions), maps, measures, models, and describes the 100 most substantive solutions to global warming". Nearly all have a component which can be impacted by innovative materials design and use.

Your Task, Part 1: Topic Selection

This is **a team assignment.** The task is aimed at exploring how careful consideration of the role of materials within a Circular Economy can play a role in maintaining and improving our lives. You are to work within a team to assess potential materials engineering solutions that have potential to contribute towards your chosen solution.

You are asked to perform an analysis of how **innovative materials selection, use and/or end-of-life processing** may be used in solving this greenhouse gas problem, and propose priority areas where materials engineering research and development or changes in materials selection could contribute to reducing greenhouse warming in your chosen area of study.

1. Examine the list of approximately 100 ranked solutions outlined at https://www.drawdown.org/solutions.
2. Nominate areas of interest from the above list via the form that is linked on Moodle.
3. You will be assigned to a team as soon as possible based on your preferences. This will likely occur towards the start of week 3 (students can still enroll in the unit up until the end of week 2)

Suggested Resources:

Videos about Drawdown
https://www.youtube.com/watch?v=u7K5M-jaHk8
https://www.youtube.com/watch?v=0zaTGMl11hs
https://www.youtube.com/watch?v=rcDA_-SZWSs
https://www.youtube.com/watch?v=v1DwrrylsrI There are numerous other similar videos online

Background Websites:
https://public.wmo.int/en/media/news/state-of-climate-report-released-un-and-wmo-chiefs https://library.wmo.int/doc_num.php?explnum_id=10211
http://www.ipcc.ch/ http://www.noaa.gov/climate www.climate.gov climate.nasa.gov
https://twitter.com/projectdrawdown?lang=en

Your Task, Part 2: Report (25%)

Write a report (15 – 20 pages of text, figures and graphs not included in limit) that contains a discussion of a proposed initiative centred on new materials engineering technologies, that may have potential to reduce greenhouse gas emissions within your topic area. Your proposed initiatives must be specific: you must describe the specific proposed location, scale, application and technology that your initiative will use.

Bad example: “We propose to replace petrochemical based plastic cutlery with biodegradable materials.”
Better example: “We propose to reprocess sugar cane waste from the XYZ sugar cane refinery in Q-Town, Queensland, turning it into poly(cutlerene) via the novel processing method described in the work of A. Scientist et al (2020), and using this material to replace 10 tonnes of polypropylene cutlery annually”

You should include:

- Discussion of the current major barriers reducing greenhouse gas emissions within your topic area
- A description of the current state of your chosen technology (if it currently exists) and how you propose that it might be applied to your specific proposal to address these barriers.
- Technical description of the production and usage of the improved material(s) in your proposed initiative, from extraction and processing through to end-of-life.
- *Quantitative* analysis (i.e. calculations, even if approximate) of the likely environmental effects of your initiative. This should include discussion of any underlying assumptions on which your analysis is highly dependent, and sensitivity analysis of these calculations.
- Analysis (quantitative (or at least semi-quantitative) where possible) of the interactions of social, political and economic factors with your proposed initiative.
- Evaluation of the initiative’s feasibility and effectiveness in improving sustainability across the triple bottom line.

Note that it is not necessary that your proposed initiative actually ends up being net beneficial; we are not expecting you to create a world-shattering new idea within the confines of this unit, and we do not expect you to abandon your initiatives and start again if you discover that they have fundamental problems that can’t be solved. In many of your assignments, the proposals will turn out to be problematic in some way; we expect you to attempt to develop them as far as possible so that they might be successful, or clearly show why they could not be successful. If you show clearly that an initiative will actually be detrimental (and show how), or show that an initiative has other problems that are not easily resolved, these are also valid and useful findings.

Your Task, Part 3: Presentation (5%)

You need to work as a group and prepare a 15 minute presentation. There will be 3-5 minutes of questions after each presentation.

Each team must submit their presentation slides (including a list of references) via Moodle by Monday 12 October at 11:55pm). The presentations are to be made during the Workshop/Practical Class in Week 11. Attendance is compulsory.

It is expected that team members work together on both the research and presentation aspects of this task. Starting from Week 3, approximately 1 hour of the practical class will be allocated for you to work together on your topic.

Generally you should:

1. Conduct your own independent literature search (including industry publications, patents, and other literature, not just research papers) on your topic and proposed materials engineering initiatives, and use this information to outline the potential role of these initiatives in greenhouse gas reduction.
2. Use the various tools discussed in weeks 1-5 of MTE4593 to analyse the likely effects of your initiatives over their life cycle, across the triple bottom line. In particular, use Granta EduPack to perform quantitative environmental life-cycle analysis where it is feasible and adds value to your discussion.
3. Provide full calculations in an appendix, but give the final results and enough information that they can be understood in context within the main report.
4. Compare the results of your calculations with any related values discussed in the literature, and discuss the sources of any discrepancies
5. Suggest any new materials properties and process solutions that may need to be developed or could assist in the future in supporting the technology for your chosen option.

Marking:

In marking your work, we will consider:

- Extent of scope: breadth of the chosen topic, and how much information is available. Teams will be rewarded for attacking topics that are complex and comprise many different aspects.
- Thoroughness of analysis: depth and attention to detail, how well you consider all the individual parts that fall within that scope, including technical, socio-political and economic considerations.
- Originality: tell us something we didn't know. Try to look at some of the less intuitive solutions in the Drawdown site, not just solutions obviously directly related to materials use, and/or you could propose some non-obvious use of materials in a particular solution scenario.
- Justification of methods: this should include a realistic assessment of uncertainties in your conclusions, and appropriate references where standard methods are used.
- Professional communication: information clearly explained for a general audience.

Moderation of team-based marks:

The team's mark will be moderated for each individual using the ITPmetrics teamwork survey, completed in week 12.

- This will provide both feedback for each student, and a quantitative "adjustment factor" that may be used to scale individual marks. This will be reserved for cases where there is a significant difference in the amount of work done by individual students.
- Students can add a confidential comment to the teaching team to explain their numerical ratings, which will be taken into account.
- Students will also be asked to provide a short comment with feedback for each other member of their team. This will be passed back to the other students as feedback on their performance.

Project management

There will also be an organised check-in during week 6, where you will be expected to discuss your progress and ideas with the teaching staff. At this stage, you should have made a reasonable start on researching the topic, and have an outline of your planned final report to discuss. An ITP Metrics survey will also be used for feedback on teamwork and monitor the team's health, so that you can implement any required changes for weeks 7-12 if needed.

Team communication:

It is expected that teams will meet synchronously on a regular basis, e.g. on-campus or via Zoom. As with all teamwork assignments, it is highly recommended that you keep written minutes for your meetings to ensure everyone agrees what was decided, to allow team members who can't attend a meeting to bring themselves up to date, and to act as evidence of the agreements that each member of the team has made. An example minutes template will be available on moodle.

To facilitate this, the second half of the practical class each week (starting from week 3) will be dedicated to the project. There will be time for your team to meet and discuss the project, and to ask questions of the staff. Teams are free to use this time (or not) as they wish; it is intended to ensure there is at least one time where the entire team can meet together each week.

Assignment of teams:

Teams will be formed by the teaching staff in order to deliberately create diverse teams, based on a range of factors. We will endeavour to place you in a team within a topic that is of interest to you. Learning to adjust and quickly get to work with others that have differing background and skillsets is an important part of being an engineer, and will be indirectly assessed during this assignment (i.e. it's not part of the marking scheme, but the teams that do it well will probably do better).